A Dual-Motor Capstan Drive Test Stand for a Quadruped Robot Leg

BDW Makers Grant — Project Report

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Figure 1: The finished test stand: two EaglePower 8308 BLDC motors mounted side by side on a 3D-printed base, each with its own Moteus r4.11 controller on the back. The cylindrical boss below the motors is the bearing housing for the shared output axle.

Abstract

With support from the BDW Makers Grant, I designed and built a test stand for a dual-motor capstan drive, the first subsystem of a 3-degree-of-freedom quadruped robot leg. Two EaglePower 8308 brushless motors each drive a small rope-wrapped drum. A 2 mm Dyneema line couples each small drum to a large output drum, giving an 8:1 reduction with no gear teeth and therefore no backlash. The two large drums share an axis but turn independently, so a single hip location can drive both input links of a planar four-bar leg. I chose the motor controllers and communication bus, sized the drums from first principles, went through four iterations of the 3D-printed base, wired and calibrated the electronics, and reached closed-loop rotation of both motors over a daisy-chained CAN-FD bus. This report covers the motivation, the design calculations, the build, the mistakes I made along the way, and what comes next.

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1 Motivation

Legged robots need actuators that are strong, fast, and *transparent*: the motor should be able to feel forces at the foot so the controller can react to impacts and uneven ground. Conventional gearboxes give torque but add backlash, friction, and reflected inertia, all of which hide those forces from the motor. Commercial quasi-direct-drive actuators solve this, but they cost far more than a student budget.

This project started from three goals:

1. **Use accessible fabrication.** Every structural part is 3D printed, so I can make complex geometry at home and change it overnight.
2. **Learn mechanisms in depth.** A robot leg brings together power transmission, kinematics, bearings, sensing, and embedded control.
3. **Iterate quickly.** Printed parts are cheap and modular, so a failed design costs a few hours instead of a machining order.

The design draws on two open-source projects: Aaed Musa's capstan drive test stand and the Stanley capstan-based quadruped kit. Both showed that a printed capstan reduction can perform close to a commercial actuator. My goal was to go past copying either design and build a *two-motor, coaxial-output* version that can drive a full leg. The total budget was about $500.

2 Background: Why a Capstan Drive?

2.1 The capstan equation

A capstan is a rotating drum with a rope wrapped around it. Sailors used them to haul anchors: a small pull on the free end of the rope holds back a very large load, because friction between the rope and the drum builds up along the whole wrap. The relationship is the capstan (Euler–Eytelwein) equation:

$$T_{\text{load}} = T_{\text{hold}}\, e^{\mu\phi}, \tag{1}$$

where T_{load} and T_{hold} are the rope tensions on the two sides of the drum, μ is the coefficient of friction between rope and drum, and ϕ is the total wrap angle in radians.

Because the holding ratio grows *exponentially* with wrap angle, a few turns of rope are enough to make slip impossible in practice. That lets a capstan drive act like a pair of gears whose "teeth" are friction distributed along the rope.

2.2 Capstan vs. cycloidal/planetary drives

Before committing, I compared the capstan drive against the cycloidal and planetary gearboxes that most hobby quadrupeds use.

Table 1: Drive mechanism trade-offs considered during planning.

	Cycloidal / planetary	**Capstan**
Backlash	Small but non-zero; grows with wear	Effectively zero when tensioned
Smoothness	Tooth engagement causes ripple	Continuous rope contact, very smooth
Torque density	High, compact	Lower; large output drum
Reduction	High ratios in one stage	Limited per stage (∼5–10:1)
Manufacturing	Tight tolerances, hard to print well	Simple cylinders and helical grooves
Maintenance	Lubrication	Rope must be kept under tension
Transparency	Moderate	Excellent (low friction, low inertia)

I chose the capstan drive because zero backlash and smooth, back-drivable motion matter most for a leg that has to sense contact. Its weaknesses (bulk and the need for tensioning) are acceptable for a test stand and can be designed around later.

3 System Architecture

3.1 Leg concept

The target leg has three degrees of freedom:

- **Two planar DOF (hip and knee):** two motors placed side by side each drive one large capstan drum. The large drums are *coaxial but independent*, so both inputs of a four-bar linkage come out of a single hip point. Offsetting the linkage attachment points on the two drums gives two independent motions from one axis of rotation.
- **Ab/adduction DOF:** a third motor rotates the whole leg assembly sideways for turning and balance.

This grant funded the core of the design: the two-motor coaxial capstan stage.

3.2 Four-bar linkage kinematics

I picked a four-bar (parallelogram) linkage for the leg because it is a closed chain, so the foot position comes straight from trigonometry. For a foot target (x, y) relative to the hip, with upper link length L_1 and lower link length L_2:

$$D = \sqrt{x^2 + y^2}, \qquad \gamma = \operatorname{atan2}(y, x), \tag{2}$$

$$\alpha = \arccos\left(\frac{L_1^2 + D^2 - L_2^2}{2L_1 D}\right), \qquad \beta = \arccos\left(\frac{L_1^2 + L_2^2 - D^2}{2L_1 L_2}\right), \tag{3}$$

$$\theta_1 = \gamma - \alpha, \qquad \theta_2 = \pi - \beta. \tag{4}$$

The forward kinematics, used to check the solution, are

$$x = L_1 \cos\theta_1 + L_2 \cos(\theta_1 + \theta_2), \tag{5}$$

$$y = L_1 \sin\theta_1 + L_2 \sin(\theta_1 + \theta_2). \tag{6}$$

Each joint angle maps to a motor angle through the drive ratio, $\phi_{\text{motor}} = N\,\theta_{\text{joint}}$, where $N = 8$ for this drive.

To build intuition before cutting any parts, I wrote an interactive 2-DOF leg simulator in React/TypeScript (available on my portfolio). It draws the linkage in SVG, runs foot trajectories in a `requestAnimationFrame` loop, and solves inverse kinematics numerically with a two-pass coarse-to-fine grid search. It also plots the commanded and actual foot paths, which made it easy to find workspace limits and singular poses.

3.3 Electronics

Table 2: Core electronic components.

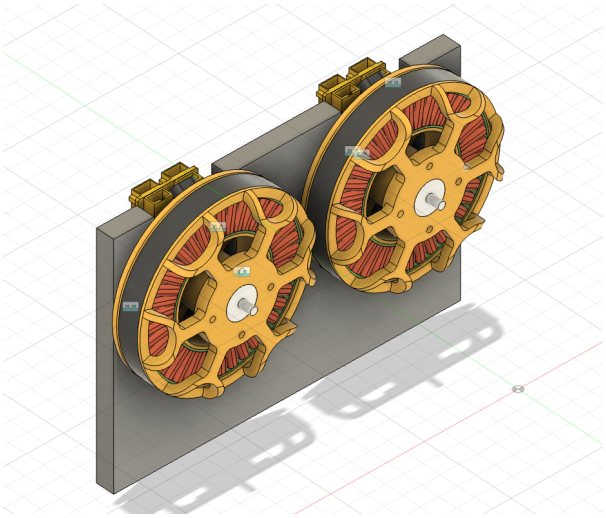
Component	**Role / key specs**
2× EaglePower 8308 BLDC (90 KV)	Actuators. 6S–12S input (22.2 V to 44.4 V), 900 W max, 22 A continuous.
2× mjbots Moteus r4.11	Field-oriented control (FOC) motor drivers, 10 V to 44 V input, onboard magnetic encoder.
mjbots fdcanusb (mjcanfd-usb-1x)	CAN-FD to USB bridge to the host computer.
24 V, 25 A AC–DC supply	Bench power for both controllers.
Diametric magnets + printed holders	Rotor position sensing for the encoders.
Raspberry Pi (planned)	Onboard host, so the robot can run untethered.

Controller choice: ODrive vs. Moteus. Aaed Musa's design uses an ODrive S1 (about $150 per controller). The Stanley quadruped uses Moteus controllers, and the r4.11 costs about $59. With a $500 budget and at least two (eventually three) motors, the Moteus was the only choice that fit. It also mounts directly on the back of the motor and reads rotor position with its onboard encoder, which saves wiring and space.

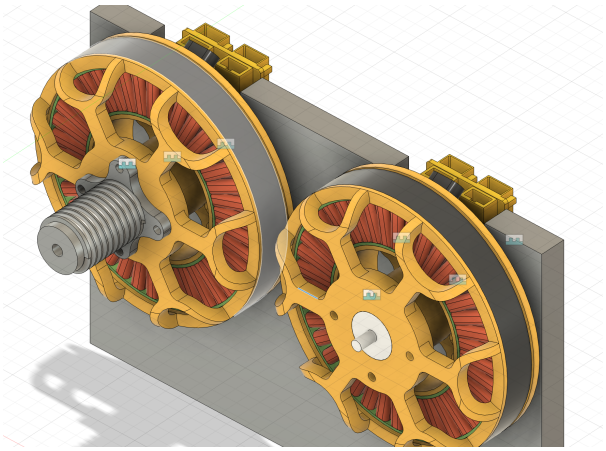
Why CAN-FD. CAN-FD (Controller Area Network with Flexible Data-rate) is a differential, multi-drop bus built for real-time control in noisy environments. All controllers share one pair of data lines, each with a unique ID (1 and 2 here). That gives three benefits:

- **Simple wiring:** controllers are daisy-chained, so there is no separate cable from each one back to the host.
- **Scalability:** a full quadruped (12 motors) only means adding nodes to the chain.
- **Autonomy later:** the host can be a laptop now and a Raspberry Pi on the robot later, with no change to the bus.

4 Capstan Drive Design



(a) Base and dual-motor layout in Fusion 360.



(b) Grooved small drum mounted on motor 1.

Figure 2: CAD of the test stand. Each motor carries a small helically grooved drum that the Dyneema line wraps around.

4.1 Setting the 8:1 ratio with effective diameters

A capstan drive's reduction comes from the ratio of drum diameters, but the diameters that count are the *effective* ones: measured at the centerline of the rope as it lies on each drum, not at the drum surface. Scaling the outer diameters by eight would give the wrong ratio, because the rope's thickness adds the same amount to both drums.

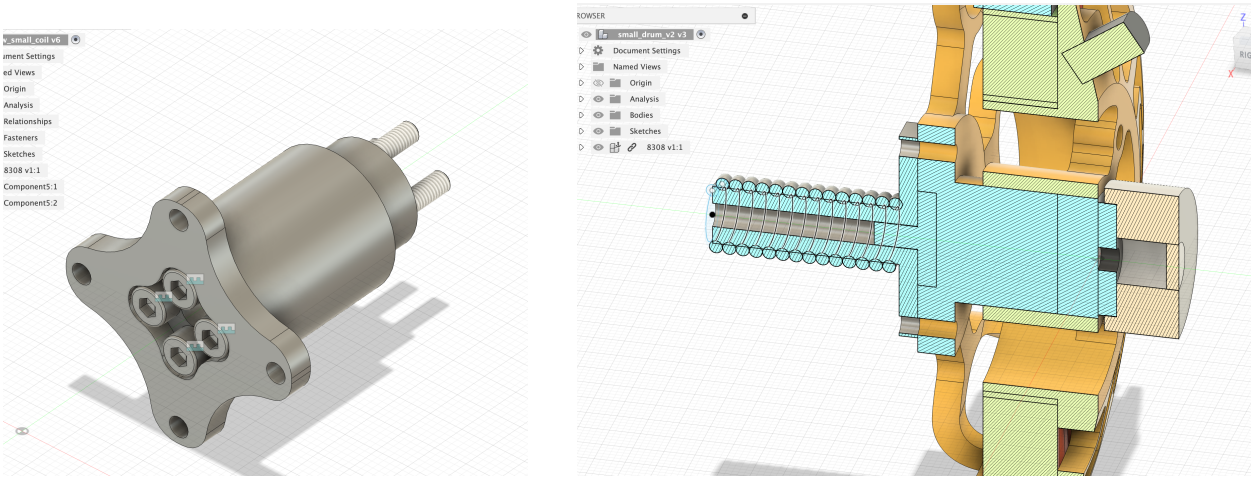
$$N = \frac{D_{\text{eff,large}}}{D_{\text{eff,small}}}. \tag{7}$$

Table 3: Drum geometry.

	Small drum	**Large drum**
Effective diameter (rope centerline)	18 mm	144 mm
Outer diameter	20 mm	146 mm
Groove coil diameter	19 mm	145 mm
Groove profile (circular section)	2.2 mm	2.2 mm
Pitch / turns / height	2.5 mm / 10 / 25 mm	
Reduction	$144/18 = 8:1$	

Bend ratio. Synthetic ropes lose strength and wear quickly when bent around too small a radius. The small drum's 20 mm outer diameter against the 2 mm Dyneema line gives $D/d = 10$, which is acceptable for Dyneema and was the limit that set the small drum size. Once the small drum is fixed, the 8:1 ratio fixes the large drum.

Grooves. Both drums have helical grooves so the rope lays down neatly and unwinds in a guided way instead of stacking or walking along the drum. In Fusion 360 I modeled each groove with the *Coil* tool (10 revolutions over 25 mm, so a 2.5 mm pitch, and a 2.2 mm circular section that leaves clearance for the 2 mm rope) and subtracted it from the drum with a *Combine–Cut*. Figure 3 shows the drum before and after.



(a) Plain drum and motor-adapter flange.

(b) Section view of the drum on the motor rotor.

Figure 3: Small drum development. The section view (b) was used to check how the drum adapter, its bolts, and the rotor fit together before printing.

4.2 Slip margin

With $\mu \approx 0.2$ for Dyneema on PLA and ten full wraps ($\phi = 10 \times 2\pi \approx 62.8\,\text{rad}$), equation (1) gives

$$\frac{T_{\text{load}}}{T_{\text{hold}}} = e^{0.2 \times 62.8} \approx e^{12.6} \approx 2.9 \times 10^5. \tag{8}$$

In operation only part of the wrap is in contact at once, so the real margin is smaller. Even so, a few wraps already give a ratio in the hundreds, so slip is not a realistic failure mode. Note that the number of wraps and the reduction ratio are *independent*: I mixed them up early on. The ratio comes only from the diameters. The wraps come from drum height and groove pitch and only set grip and contact area.

4.3 Expected output performance

Using the motor data in Table 2, a first-order estimate of the drive output (ignoring rope and bearing losses) is:

- **Torque constant:** $K_t \approx 8.27/K_V = 8.27/90 \approx 0.092\,\text{N m/A}$.
- **Continuous output torque:** $0.092 \times 22\,\text{A} \times 8 \approx 16\,\text{N m}$ at the large drum.
- **No-load output speed at 24 V:** $90 \times 24/8 \approx 270\,\text{rpm}$.
- **Rope tension at that torque:** $16\,\text{N m}/0.072\,\text{m} \approx 220\,\text{N}$, far below the breaking strength of 2 mm Dyneema (several kilonewtons).

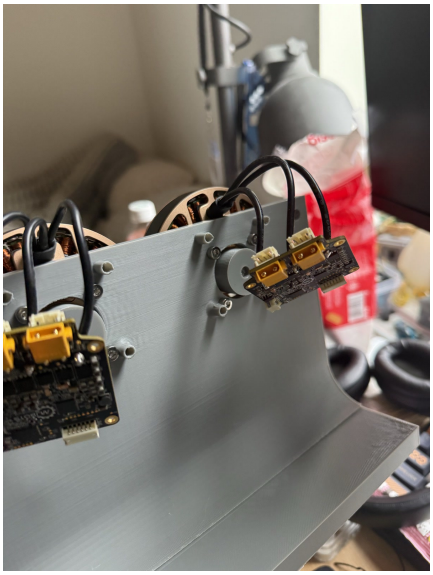
The 8:1 ratio keeps the drive *quasi-direct*: high enough to multiply torque and low enough that the leg stays back-drivable and the motor current remains a good proxy for foot force.

5 Mechanical Design Iterations

The first layout put two motors on a flat plate. A major revision on 14 November reorganized the base around the coaxial output: it now carries the coaxial drum plus a "through" drum and an axial drum that supports the through-axle, so the two output drums share an axis while turning independently.

The base went through **four printed iterations**. The main changes:

- **Bearings:** 15×28×7 mm bearings for the small drum shafts and 42×50×12 mm thin-section bearings for the large coaxial axle.
- **Fasteners:** heat-set inserts with M2.5 screws for the Moteus boards (so they can be removed repeatedly without wearing out the plastic), and M3 screws for everything else.
- **Encoder magnet holder:** a dedicated printed holder keeps the diametric magnet centered on the motor axis behind each controller.
- **Controller placement:** the Moteus mounting holes were in the wrong place on an earlier base. This version fixes that so the onboard encoder sits directly over the magnet.



(a) Back of the base with both Moteus r4.11 boards mounted on heat-set inserts over the encoder magnets.



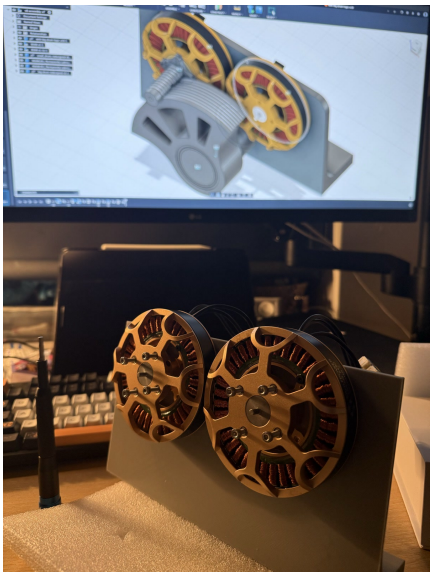
(b) Thin-section bearings for the coaxial output axle.

Figure 4: Printed base and bearings.

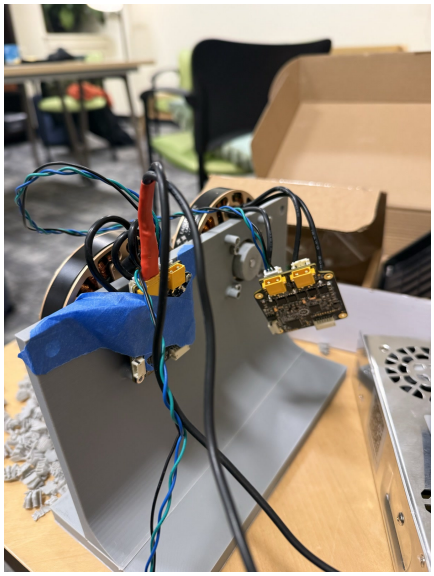
6 Build and Bring-Up

6.1 Assembly

All parts arrived in early November. After the base was printed, I screwed in the motors and controllers, pressed in the encoder magnets, and soldered XT30 connectors to the power leads (Figure 5).



(a) Motors mounted on the printed base, next to the CAD model.



(b) Power and CAN wiring on the back of the stand.

Figure 5: Assembly of the test stand.

6.2 First power-on and calibration

On the first power-on I reversed the supply polarity and the board began to smoke. I cut power immediately. With the polarity corrected, the controller came up and its status LED blinked normally. The first calibration then failed because of a bad configuration. After fixing it, with help from Josh Pieper (the Moteus developer) on the mjbots Discord, the motor completed calibration and spun under closed-loop control for the first time on 23 November (Figures 6–8).

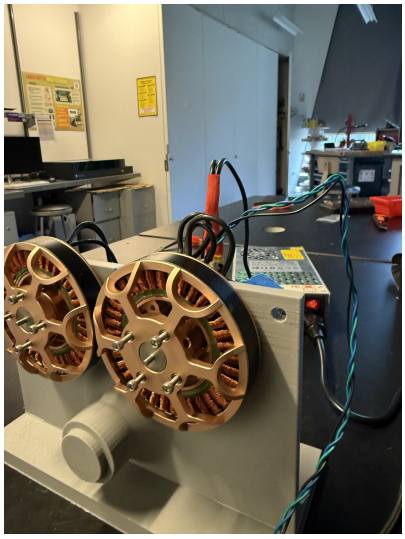


Figure 6: Both motors and controllers wired to the 24 V supply for the first power-on.

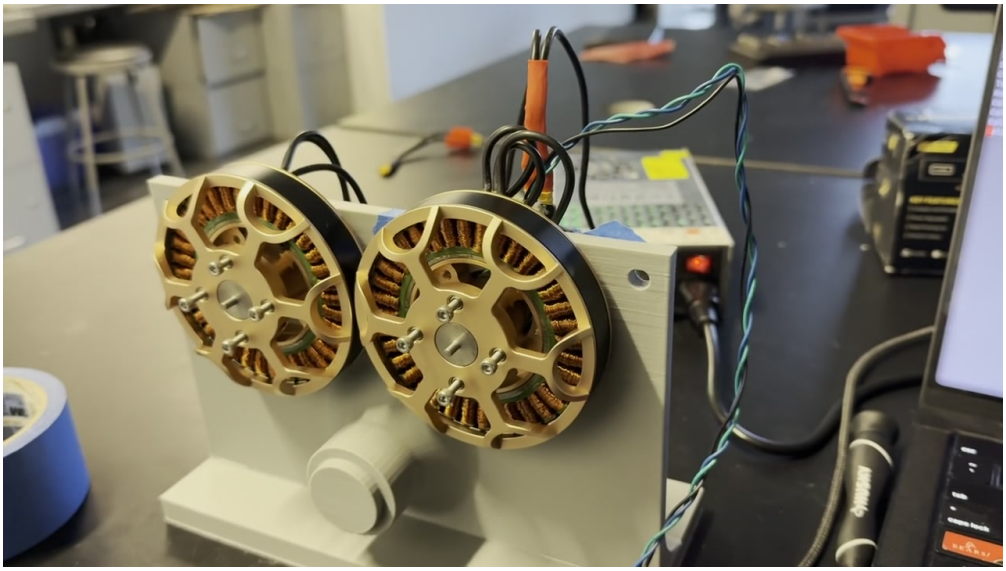
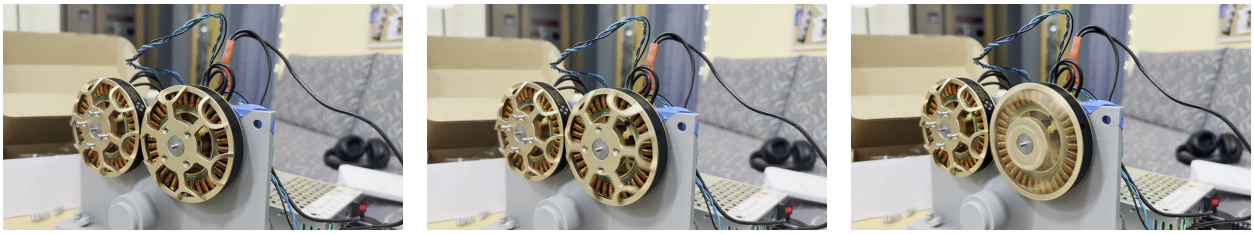


Figure 7: Frame from the calibration video. During calibration the Moteus steps the motor through electrical cycles to measure the encoder offset, pole count, phase resistance and inductance.



(a) At rest. (b) Spinning up. (c) At speed.

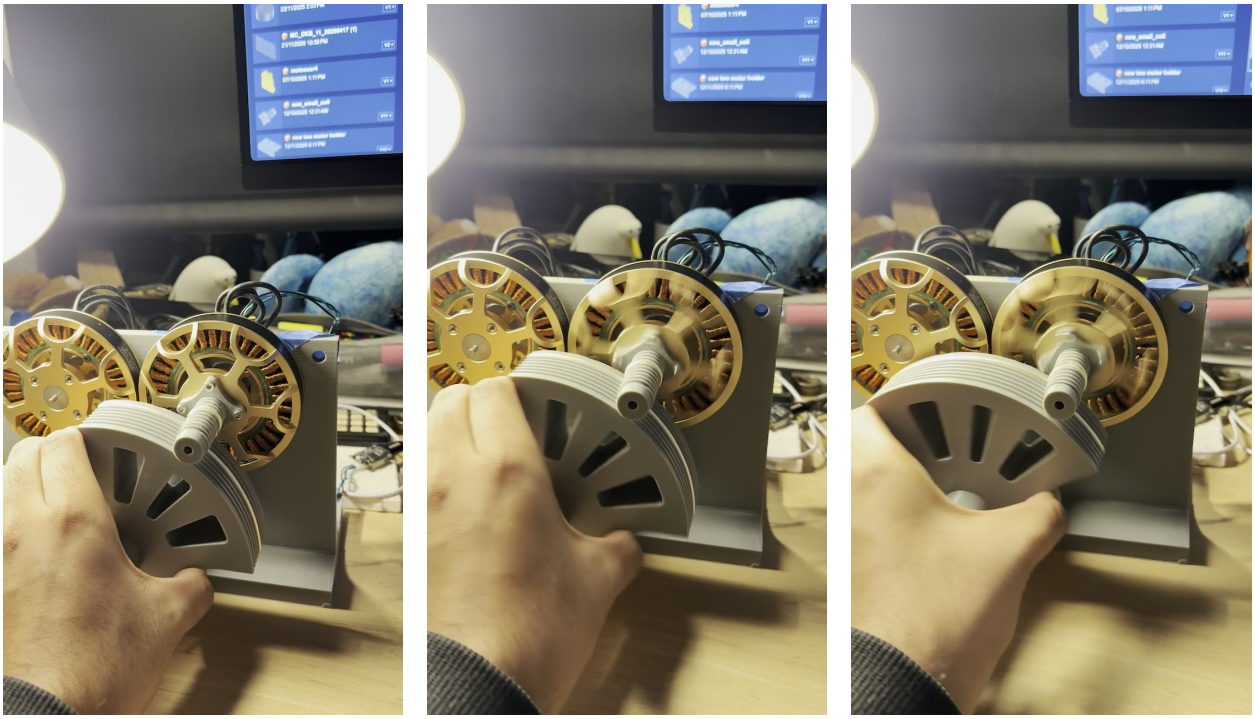
Figure 8: Frames from the first successful rotation. The right motor, under closed-loop control, blurs as it spins while the left motor stays still.

6.3 Two-motor control

Next I set up the two controllers on one CAN-FD bus with IDs 1 and 2, powered in parallel from the 24 V supply, and first checked them with `tview`. I then wrote small Python scripts on top of the `moteus` asyncio library. The main one, `dual_pos.py`, takes position targets for both motors (in turns), sends trajectory-limited position commands (velocity 0.8 turn/s, acceleration 2 turn/s^2, torque limit 2 N m) with a watchdog timeout, and refreshes the command until each move finishes. Other scripts cover sweeps, keyboard and arrow-key jogging, and linkage calibration.

6.4 Hand test of the capstan stage

Before running the drive under power, I wrapped a printed large drum and turned it by hand (Figure 9). The 8:1 reduction is easy to see: a small movement of the large drum spins the motor rotor through a much larger angle. The motion was smooth, with no noticeable backlash when reversing direction.



(a) (b) (c)

Figure 9: Back-driving the capstan stage by hand. The grooved large-drum sector turns about the output axle, and the rope drives the small threaded drum on the right-hand motor. Because the drive is back-drivable, the motor can be turned from the output side, which is what makes it suitable for force-sensitive legged control.

7 Lessons Learned

- **Check polarity before the first power-on.** Reversed supply leads nearly cost me a controller. I now check polarity with a multimeter and use keyed XT30 connectors throughout.
- **Use color-coded wire.** An all-black daisy-chain harness looked clean but caused a soldering mistake. Red/black power and colored CAN pairs are worth giving up the looks.
- **Ratio and wraps are separate design choices.** Diameters set the reduction; wraps set grip.
- **Mind the encoder air gap.** My first magnet holder put the magnet less than 1 mm from the sensor, against a recommended gap of about 2 mm. It worked, but it is being redesigned for reliable readings.
- **Confirm part dimensions on arrival.** One bearing order arrived in the wrong size and had to be returned, which held up assembly.
- **Ask the community.** The Moteus developer's help on Discord turned a stalled calibration into a working motor in an afternoon.

8 Next Steps

1. Print the second motor's capstan drum and the main plate, then rope and tension both stages.
2. Measure the real reduction ratio and adjust the large drum diameter to account for how much the rope flattens under tension.
3. Redesign the encoder magnet holder for a 2 mm gap.
4. Characterize the drive: backlash, back-drive torque, torque ripple, and how much tension is lost over time.
5. Attach the four-bar linkage and run the simulator's trajectories on hardware.
6. Add the ab/adduction motor and move control onto a Raspberry Pi for an untethered 3-DOF leg, the first step toward a full quadruped.

Acknowledgments

Thank you to the BDW Makers Grant for funding this project, to Aaed Musa and the Stanley quadruped project for publishing their designs openly, and to Josh Pieper of mjbots for his help with Moteus configuration.

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